

Coexistence of virulence and β -lactamase genes in avian pathogenic *Escherichia coli*

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ABSTRACT

Emergence of multidrug resistance in *E. coli* and advent of newer strains is becoming serious concern which requires keen observations. This study was designed to find the ciprofloxacin resistant *E. coli* isolates co-existed with multi-drug resistance along with β -lactamase production from poultry source, and finally the genome sequencing of these strains to explore genetic variations. Study constituted on isolation of $n = 225$ *E. coli* from broiler farms of central China which were further subjected to identification of resistance against ciprofloxacin followed by antibiogram of $n = 26$ antibiotics and identification of β -lactamase production. Whole genome resequencing was performed using Illumina HiSeq 4000 system. PCR results revealed predominant β -lactamase genes i.e. CTX-M, CTX-M-1, CTX-M3, TEM-1 and OXA. Furthermore, the MDR isolates were containing most of the tested virulence genes. The most prevalent virulence genes were *pap-C*, *fim-C*, *fim-H*, *iuc-D*, *irp-2*, *tra-T*, *iro-N* and *iut-A*. The single nucleotide polymorphisms (SNPs) loci mentioned in this data give valuable genetic markers to growing high-throughput techniques for fine-determination of genotyping of MDR and virulent isolates. Characterization of SNPs on functional basis shed new bits of knowledge on the evolution, disease transmission and pathogenesis of MDR *E. coli* isolates. In conclusion, these findings provide evidence that most of poultry *E. coli* are MDR, β -lactamase producers, and virulent which could be a zoonotic threat to the humans. The whole genome resequencing data provide higher resolution of resistance and virulence characteristics in *E. coli* which can further be used for the development of prevention and treatment strategies.

1. Introduction

Antibiotics strengthen the present medicines as their usage has decreased death rate, increased quality life span, and are vital for chemotherapy and chemosurgery. However, infections due to multidrug-resistant (MDR) bacteria are increasing worldwide, leading to increased numbers of untreatable diseases. Undoubtedly, non-judicious use of antimicrobials not only aggravates resistance but also disturbs

normal microflora of treated animals [1,2]. Antimicrobial resistance has appeared as international issue that is significantly compromising health of animals and humans where *E. coli* and *S. aureus* cannot be ignored [3–7].

Global spread of pathological and MDR bacteria is persistently decreasing the efficacy of existing antibiotics while the new and under production antibiotics are the derivatives of existing antibiotics. Therefore, more chances of cross-resistance may decrease the efficiency

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of new antibiotics. It is estimated that antimicrobial resistance will cause almost 10 million deaths annually by 2050 [8]. Among several drugs are the beta-lactam drugs that have become significant to receive resistance by pathogens. Extended-spectrum β -lactamases (ESBLs) producing bacteria have now become resistant to the drugs other than first, second and third generation cephalosporins. Therefore, there is a quest for understanding the resistance mechanism and to develop substitute antibiotic therapy [9,10]. *E. coli* is versatile bacteria that is commensal in some cases while turn into pathogenic both in humans and animals resultantly giving rise to wide range of health issues [11].

Existence of resistance genes have been studied not only in poultry but also in other food producing animals, pets, and humans [12,13]. Many researchers discuss potential transmission of *E. coli* clones from animals to human, and vice versa [12,14–16], and contaminated food played an important role in the transmission of resistant *E. coli* isolates [17]. Additionally, a strong relationship has developed between poultry and human MDR *E. coli* due to sharing of similar genetic characteristics [18]. Some of the *E. coli* spp. harbor certain ESBLs (CTX-M group) that are becoming significant threat to the health throughout the world by inhibiting beta-lactam antibiotics [19], thus limiting treatment options for infections [20]. As a specie, avian pathogenic *E. coli* (APEC) has been known to trigger different pathogenic phenotypes to enhance serious infections in poultry [21]. Jackson and his co-workers identified that about 92% (180/195) of poultry meat samples were contaminated with *E. coli*, while 46% of samples owned virulence genes related [22].

Virulent determinants of pathogen help to evade host immune response while development of antimicrobial resistance empower bacteria against overwhelmed antibiotic treatments and to adjust in unfavorable conditions [23]. Virulence genes of pathogen generally play their role in adhesion, invasion, toxicity, immune suppression, iron acquisition and bacterial survival [24]. Since intensified production has been generally connected with the poultry industry and many outcomes demonstrated that antibiotic resistance and virulence of *E. coli* were very serious issues. Therefore, control of APEC has been an incredible task [25,26].

It is usual to execute PCR or micro-arrays to identify resistant genes in pathogen, but authenticity requires DNA sequencing that in turn if compared with conventional methods becomes cost and time effective. Therefore DNA sequencing is becoming more popular and easily available for consistent use for the detection of virulence and resistance characteristics from microbes [27]. Whole-genome resequencing (WGR) involves sequencing the entire genome of a bacteria, and comparing the sequence to that of a known reference, providing a lot of information about antimicrobial resistance and virulence factors, which can further be used as an alternative of commonly used laboratory testing for investigation of antimicrobial resistance and the detection of virulence and resistance phenotypes [28,29]. The hypothesis of this research thus states as “ciprofloxacin resistant *E. coli* are prevalent in poultry, and there is co-existence of multidrug resistant genes along with it.

2. Materials and methods

2.1. Samples collection

For clinical pathogenic *E. coli*, $n = 900$ fecal samples of broilers showing signs of colibacillosis and having history of use of different antibiotics (some of antibiotic used in the study area were penicillin, amoxicillin, erythromycin, tylosin, bacitracin, chlortetracycline, oxytetracycline, etc.) were collected from 25 different commercial poultry farms of Henan, Hubei, and Hunan provinces of China between March 2015 and July 2016. For the primary isolation of *E. coli*, samples were cultured in Mueller-Hinton (MH) broth and then inoculated on MacConkey agar medium and incubated at 37 °C for 24 h. A dark pink colony was randomly selected from each plate and subculture on EMB (Eosin Methylene Blue) agar plates. One blue-black colony (presumptive *E. coli*) with metallic sheen growing on EMB plates was selected for

further analysis. Total 225 *E. coli* isolates were selected. All isolates were identified as *E. coli* based on their colony morphology on EMB agar and subsequent biochemical testing [30]. Additionally, all *E. coli* isolates were then confirmed biochemically using the API 20E system (Bio-Mérieux, Marcy-l'Étoile, France). The isolates were confirmed as avian pathogenic *E. coli* by multiplex PCR as previously described [31]. Total 45 ciprofloxacin resistant (≥ 16 mg/mL) *E. coli* isolates were further found which were proceeded for onward study. The quality control *E. coli* ATCC 25922 obtained from Chinese Veterinary Culture Collection, was used as reference strain. All *E. coli* isolates were stored at -80 °C until further use. Before performing each experiment, isolates were first grown in Mueller-Hinton (MH) broth overnight and then on EMB agar with incubation conditions of 37 °C for 18–24 h [33].

2.2. Antimicrobial susceptibility test

Collected isolates along with reference strain (ATCC 25922 *E. coli*) were cultured on selective MacConkey agar and subjected to adjusting growth of bacteria at 5×10^5 CFU/mL. The adjusted growth was used in broth microdilution method against ciprofloxacin using two fold dilution method as per prescribed by CLSI guidelines [32] to find minimum inhibitory concentration (MIC). Briefly describing the protocol, 96 wells titration plate was used to dilute antibiotics by two folds and later culture was added, and titration plate was incubated for 24 h at 37 °C. Optical density (OD) value of each well before and after incubation was measured to find effect of antibiotics' dilution on growth of bacteria by comparing net OD value with cut off value. The isolates resistant to ciprofloxacin were further subjected to calculation of MICs of different antibiotics [33].

2.3. β -Lactamase identification

All *E. coli* isolates along with reference strain *E. coli* ATCC 25922 were screened for β -lactamases, and then phenotypic confirmatory test was performed as per CLSI protocol. According to the CLSI protocol, ceftazidime and cefotaxime antibiotic-impregnated paper disks were used for screening of ESBL containing strains by Kirby–Bauer disk diffusion method. More than one antibiotic disks were used to improve the sensitivity of β -lactamase detection, as suggested by CLSI guidelines [32]. Inoculum with concentration equivalent to 0.5 McFarland standard was prepared from colonies on MH agar plates and spread by lawn culture method with sterile cotton swabs. Using sterile forceps, antibiotic disks were placed on these plates and were incubated aerobically at 37 °C for 18–24 h. ATCC 25922 *E. coli* and ATCC 700603 *Klebsiella pneumoniae* were used as negative and positive control strains respectively, for β -Lactamase production [33].

2.4. Phenotypic confirmation of β -lactamase production

Phenotypic confirmation test for β -lactamase producers was done by double-disks synergy method. For this test, ceftazidime (30 μ g) or cefotaxime (30 μ g) disks with or without clavulanic acid (10 μ g) were used, as per CLSI guidelines. Lawn cultures of susceptible isolates were made on MH agar and disks were placed at distance of 30 mm from center to center. Plates were aerobically incubated at 37 °C for overnight. The difference in inhibition zone ≥ 5 mm of cephalosporin disks and their clavulanate combination indicated the ESBL production [32]. The phenotypic confirmation was further validated through broth microdilution method. Briefly, cefotaxime along with clavulanate (0.25/4 to 64/4 μ g/mL); cefotaxime alone (0.25 64 g/mL); ceftazidime with clavulanic acid (128 mg/mL); and ceftazidime alone (128 mg/mL) were used. Reduction in MIC by three-to-two-fold serial dilution was set as basis for phenotypic confirmation. Standard protocols as described in CLSI were followed for this experiment [32].

2.5. PCR amplification of β -lactamase genes

For further molecular analysis, ciprofloxacin resistant *E. coli* co-existed with multidrug resistance and phenotypically positive for β -Lactamase were selected. These were 6 such kind of isolates representatively selected for PCR assay which was performed using different starting primers as described previously in different studies (Table 1).

MDR *E. coli* isolates along with reference strain ATCC 25922 *E. coli* (β -lactamase negative control) were cultured in MH broth at 37 °C overnight and then centrifuged. Following the instructions systematically, sediments were used for the extraction of DNA using DNA kit (TianGen Biotech. Co. Ltd. China). Extracted DNA was used as the templates for PCR amplification. The PCR mixture consist of 1 μ L from each part of primer, 4 μ L DNA template, 10 μ L 2X PCR Master Mix (containing 2X dNTP PCR mixture, 2X PCR buffer, 2X Taq DNA polymerase) (TaKaRa Biotech. Co. Ltd. China) and deionized water to the final volume of 25 μ L [46]. DNA Marker DL 2000 was used (TaKaRa Biotechnology (Dalian) Co. Ltd.). PCR was completed following different conditions as, first denaturation at 95 °C for 5 min, second denaturation at 94 °C for 30 s, annealing at 53–60 °C for 30 s, elongation at 72 °C for 60 s with 35 rounds of cycles and then final elongation at 72 for 10 min. PCR products (10–15 μ L of each) were evaluated by agarose gel electrophoresis at 100 V and 130 mA for 45–60 min, in 1.2% agarose gel prepared in 0.5 $\times$ Tris-borate-EDTA (TBE) buffer having stain ethidium bromide (1 mg/mL) and imagined on a UV transilluminator.

2.6. Identification of virulence genes

To show the virulence characters of the selected isolates, nineteen most prevalent virulent genes in poultry *E. coli* were checked by PCR assay. Primers of these genes are given Table 2 with little modification [47].

Table 1

Oligonucleotides used for PCR amplification of β -lactamase genes in poultry *E. coli*.

Genes	Forward/Reverse Primers	Gene primer sequence (5'-3')	Size (bp)	References
CTX-M	CTX-M F CTX-M R	CGCTTTGCGATGTGCAG ACCGCGATATCGTTGGT	550	[34]
CTX-M-1	CTX-M-1 F CTX-M-1 R	GCTGTTGTAGGAAGTGTGCCGC GCCGCCGACGCTAATACATC	798	[35]
CTX-M-2	CTX-M-2 F CTX-M-2 R	AAATGTGCTGCTCTTTCTGTGAGC AGGGTTCGTGCAAGACAAGACTG	1122	[36]
CTX-M-3	CTX-M-3 F CTX-M-3 R	GTTACAATGTGTGATAAGCAG CCGTTTCCGCTATTACAAAC	800	[37,38]
OXY	OXY F OXY R	GGTTTGGTAACTGTGACGGG CAGAGTGCAGAGTGTTCAG	725	[39]
OXA	OXA F OXA R	TCAACTTTCAAGATCGCA GTGTGTTTGAATGGTGA	591	[40,41]
TEM-1	TEM-1 F TEM-1 R	ATAAAATCTTGAAGACGAAA GACAGTTACCAATGCTTAATC	1080	[40,41]
CMY	CMY F CMY R	GATTCTTGGACTCTTCAG TAAACCCAGGTTCCAGATAGC	1807	[42]
CMY-2	CMY-2 F CMY-2 R	CTGCGCTCTGCTGCTGACAGC CTCGACACGGACAGGGTTAGG	1093	[43]
SHV	SHV F SHV R	CCCTGTTAGCCACCCTGCCG CGTTGCCAGTGTCTGATCAGC	829	[43]
DHA	DHA F DHA R	AACCTTACAGGTGTGCTGGGT CCGTACGCATACTGGCTTTGC	405	[44]
NDM	NDM (NDM-1) F NDM (NDM-1) R	GGTTTGGCGATCTGTTTTC CGGAATGGCTCATCAGCATC	621	[45]
KPC	KPC F KPC R	CGTCTAGTTCTGCTGTCTTG CTTGTCATCCTTGTAGGCG	798	[45]

2.7. Whole genome Re-sequencing

Ciprofloxacin resistant *E. coli* isolates resistant to multidrug and β -lactamase producers were selected for next generation sequencing to study deep into genetic analysis. All isolates were cultured in MH broth, centrifuged at their log phase while DNA were extracted using DNA Kit following the manufacturer's instructions. Samples were shipped to the company, "Beijing Genomics Institute (Shenzhen, China)" for resequencing of all isolates. The samples were tested, the library was constructed using the test sample, and the cluster preparation and sequencing with qualified libraries. The assembly results of scaffold and reference sequences (*Escherichia coli* DH1, complete genome, GenBank: CP001637.1, <https://www.ncbi.nlm.nih.gov/nuccore/260447279>) were compared for SNP detection, insertion loss (InDel) detection, and respectively annotate SNPs and InDels. Phylogenetic tree based on SNP matrices of samples and reference strains was prepared.

SOAPdenovo was used to assemble short sequence assembly software clean data [56,57]. After several adjustments excellent assembly results were obtained. Using SOAPaligner (version: 2.21) sequence alignment software, all of the sequenced read sequences and reference sequences were aligned [58,59]. Based on the results of the comparison, the average depth and coverage were calculated. SNPs detection and annotation based on assembly results and reference sequence alignment were performed. Using MUMmer alignment software global gene alignment [60], each sample with the reference sequence was used to find the query sequence and reference sites and carried out a preliminary filtering and detection of potential SNP sites. The 100 bp sequence fragment of each SNP of the reference sequence was extracted, and the extracted sequence and assembly results were then subjected to BLAST. The duplicates SNPs and SNPs with ≤ 101 bp were considered untrustworthy and were removed. Repeated sequence sites were predicted using BLAST, TRF [61] and Repeatmask [62] softwares, the small SNPs were removed by filtration and finally, got reliable SNPs.

Phylogenetic tree based on SNP matrices of samples and reference strains was constructed to check the evolution of species. PHYML algorithm of TreeBeST [63] was used to develop a phylogenetic tree whereas the bootstraps parameter was set to 1000. Identification of InDels in light of next generation sequencing (NGS) advances is ending up more typical, and current methodologies like split reads, gapped arrangement and *de novo* assembly [64]. The 150bp (3*SD) in the upstream and downstream of InDel site in the reference sequence were extracted and then aligned with the query reads. The alignment results were verified with BWA (http://bio_bwa.sourceforge.net/) and smtools (<http://samtools.sourceforge.net/>). The genome sequence data files have been submitted to BioProject database of NCBI and can be accessed through accession number PRJNA768944. Bioinformatics tools like ResFinder and VirFinder, were applied used to identify antimicrobial resistance and virulence genes. The similarity cutoff or identity threshold for reporting a match between a gene in the resistance finder database (RFDDB) or virulence finder database (VFDB) and the input microbial genome was set at $\geq 90\%$ identity over $\geq 60\%$ of the length of the target gene [65].

3. Results

3.1. Antibiotic susceptibility test

Ciprofloxacin (CPF) resistant *E. coli* isolates were found further resistant against cefotaxime by 75.55% followed by imipenem (66.66%) and enrofloxacin (62.22%). None of isolate these isolates were found resistant against spectinomycin. It was observed that 15.55, 13.33 and 11.11% isolates were resistant to 6, 7 and 8 types of antibiotics, respectively. Two isolates were resistant to 20 and 17 antibiotics, respectively, indicating emerging extensive drug resistance in *E. coli*. The results of susceptibility testing are presented in Fig. 1. The MIC values of the selected *E. coli* isolates along with reference strain ATCC

Table 2Oligonucleotides used for PCR amplification of virulent genes in poultry *E. coli*.

Genes	Forward/Reverse primers	Primers	Size (bp)	References
O157	O157 F O157 R	GTAGGGAAGCGAACAGAG AAGCTCCGTGTGCCTGAA	361	[48]
papC	papC F papC R	GACGGCTGTACTGCAGGGTGTGGC ATATCCTTTCTGCAGGGATGCAATA	328	[49]
hlyE	hly-E F hly-E R	TCGGCATCCACATTAGTTG AATCGAGTTGTTTCCGTCTCT	384	[48,50]
stx-1	stx-1 F stx-1 R	TCTCAGTGGGCGTTCTTATG TACCCCTCAACTGCTAATA	338	[48,50]
Stx-2	Stx-2 F Stx-2 R	GCGGTTTTATTGCTTAGC TCCCGTCAACCTTCACTGTA	115	[48,50]
eaeA	eaeA F eaeA R	ATGCTTAGTGCTGGTTAGG GCCTTCATCATTTTCGCTTTC	248	[48,50]
tsh	tsh F tsh R	ACTATTCTCTGCAGGAAGTC CCTCCGATGTTCTGAACGT	825	[51]
Vat	vat F vat R	TCCTGGGACATAAYGGTCAG GTGTCAGAACGGAATTGT	981	[47]
irp-2	irp-2 F irp-2 R	AAGGATTCGCTGTACCGGAC TCGTCCGGCAGCGTTTCTTCT	280	[47]
hlyF	hlyF F hlyF R	GGCCACAGTCGTTTAGGGTGCTTACC GGCGGTTTAGGCATTCCGATACTCAG	599	[47]
iucD	iucD F iucD R	ACAAAAAGTTCTATCGCTTCC CCTGATCCAGATGATGCTC	714	[47]
iroN	iroN F iroN R	AAGTCAAAGCAGGGTTGCCCG GACGCCGACATTAAGACGCAG	667	[47]
fyuA	fyuA F fyuA R	GCGACGGGAAGGGATGACTTA CGCAGTAGGCACGATGTTGTA	949	[47]
ompT	ompT F fimH F	TCATCCCGAAGCCTCCCTCACTACTATTA CGAGTTATTACCTGTTTGCTG	559	[47]
fimC	fimC F fimC R	GGGTAGAAAATGCCGATGGTG CGTCATTTTGGGGTAAGTGC	508	[47]
fimH	fimH F fimH R	CGAGTTATTACCTGTTTGCTG ACGCCAATAATCGATTGCAC	878	[52]
iss	iss F iss R	ATGCAGGATAATAAGATGAAA CTATTGTGAGCAATATACA	290	[53]
iutA	iutA F iutA R	GGCTGGACATCATGGGAAGTGG CGTCGGGAACGGGTAGAATCG	302	[54]
traT	traT F traT R	GGTGTGGTCCGATGAGCACAG CACGGTTCAGCCATCCCTGAG	290	[55]

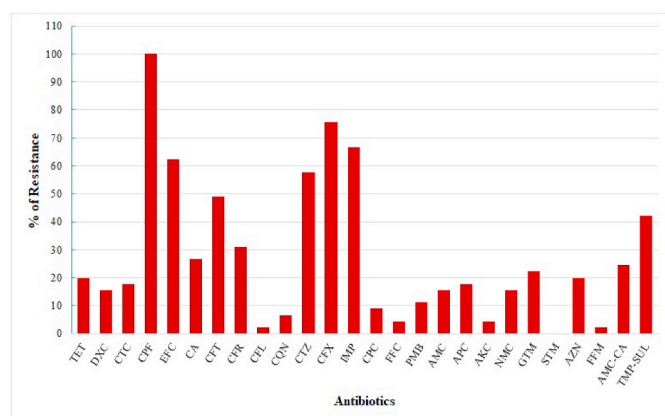


Fig. 1. Rate of antibiotic resistance in *E. coli* isolated from poultry. The prevalence of different antibiotics is presented in the percentage (%) of isolates. [Tetracycline (TET), doxycycline (DXC), chlortetracycline (CTC), ciprofloxacin (CPF), enrofloxacin (EFC), clavulanic acid (CA), ceftriaxone (CFT), ceftiofur (CFR), cefalothin (CFL), cefquinome (CQN), ceftazidime (CTZ), cefotaxime (CFX), imipenem (IMP), chloramphenicol (CPC), florfenicol (FFC), polymyxin B (PMB), amoxicillin (AMC), ampicillin (APC), amikacin (AKC), neomycin (NMC), gentamicin (GTM), spectinomycin (STM), aztreonam (AZN), fosfomycin (FFM), amoxicillin-clavulanic acid (AMC-CA), and trimethoprim-sulfonamide (TMP-SUL)].

25922 against 26 antibiotics are given in [Supplementary Table S1](#).

3.2. β -Lactamase identification

Isolates with less zone diameter against respective antibiotic (ceftazidime zone ≤ 22 mm and cefotaxime zone ≤ 27) were categorized as positive strains isolates, mean β -Lactamase producers. The study found fewer isolates resistant to ceftazidime or cefotaxime, but they were not β -lactamase positive isolates. On the other hands, most of the isolates resistant to ceftazidime were also resistant to cefotaxime.

3.3. Phenotypic confirmation of β -lactamase production

We confirmed β -lactamase production by double-disk synergy test and then verified by microdilution method and found that out of 45 *E. coli* isolates, 25 (55.5%) were β -lactamase producers. The antimicrobial resistance and MDR rates were significantly higher in β -lactamase positive isolates as compared to β -lactamase negative isolates ([Fig. 2](#)).

3.4. Confirmation of β -lactamase genes

From a total of 13 β -lactamase genes, isolates 351, 357 and 381 contained 5 genes while isolates 112 and 130 contained 4 genes and isolate 27 harbored only 3 genes. The prevalence of investigated β -lactamase genes in selected *E. coli* isolates is given in [Table 3](#).

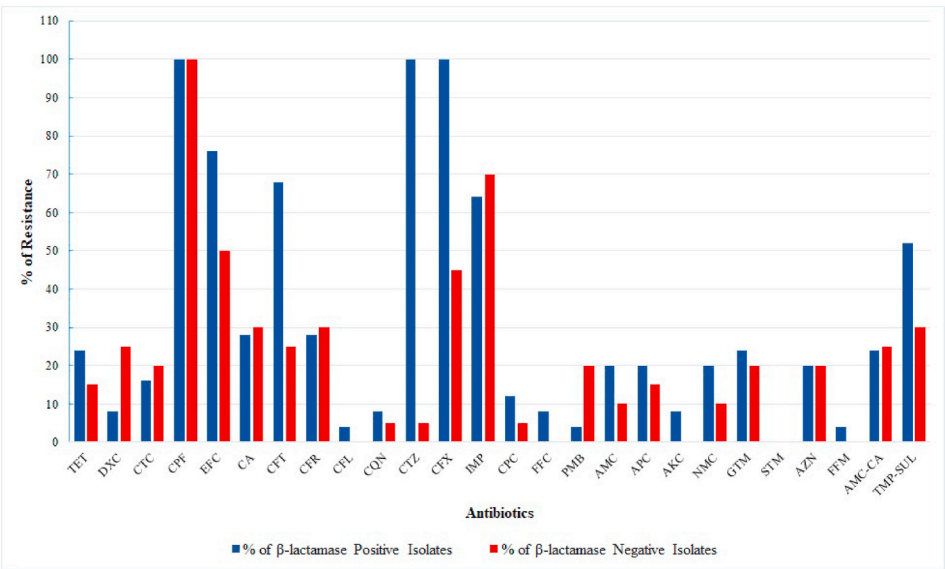


Fig. 2. Comparison of antibiotic resistance rates in β -lactamase positive and β -lactamase negative *E. coli* of poultry are presented in percentage of isolates.

Table 3
Selected clinical isolates of poultry *E. coli* producing β -lactamases genes.

<i>E. coli</i> Isolates	β -Lactamase genes Identified
27	CTX-M, CTX-M-1, CTX-M3
112	CTX-M, TEM-1, OXA, OXY
130	CTX-M, CTX-M-1, CTX-M3, OXA
351	CTX-M, CTX-M-1, CTX-M3, OXA, OXY
357	CTX-M, CTX-M-1, CTX-M3, OXA, OXY
381	CTX-M, CTX-M-1, CTX-M3, TEM-1, OXA

3.5. Virulence genes identification by PCR

Out of 19 checked virulence gens, genes *hly-E*, *hly-F*, *stx-2*, *eae-A*, *tsh*, *vat*, *iss* and *fyu-A* were not present in any isolate, while 6 genes *pap-C* (P-fimbriae), *fim-C* (type-1 fimbriae), *fim-H* (adhesion), *iuc-D* (aerobactin), *tra-T* (serum resistance) and *iut-A* (aerobactin siderophore receptor) were present in isolates. Results of virulence genes prevalence are summarized in Table 4.

3.6. Whole genome Re-sequencing

The graphical presentation of InDel in given in Fig. 3. We used TreeBeST to build phylogenetic tree. On the bases of SNP analysis results between isolates, phylogenetic tree was constructed (Fig. 4). The statistics of illumina sequencing are shown in Table 5.

There were a very large number of SNPs present in tested *E. coli* isolates with respect to the reference strain *E. coli* DH1, which were involved in lot of enzymes and protein production. By “Short Oligonucleotide Analysis Package” we found that total 4160 reference genes involved in SNPs of our tested isolates. Some important SNPs are given in Table 6.

Table 4
Selected clinical isolates of *E. coli* producing virulence genes.

<i>E. coli</i> isolates	Virulence genes Identified
27	pap-C, fim-C, fim-H, iuc-D, tra-T, iut-A
112	O157, pap-C, fim-C, fim-H, iuc-D, tra-T, opm-T, iut-A
130	pap-C, fim-C, fim-H, iuc-D, tra-T, iro-N, iut-A
351	O157, pap-C, fim-C, fim-H, iuc-D, tra-T, iut-A
357	pap-C, fim-C, fim-H, iuc-D, tra-T, iro-N, iut-A
381	pap-C, fim-C, fim-H, iuc-D, irp-2, tra-T, iro-N, iut-A

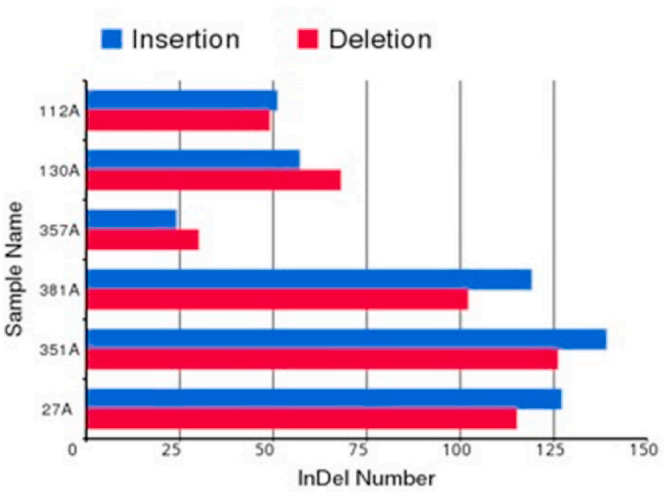


Fig. 3. Graphical representation of InDel length distribution. The X-axis is InDel length and the Y-axis is sample Insertion and Deletion.

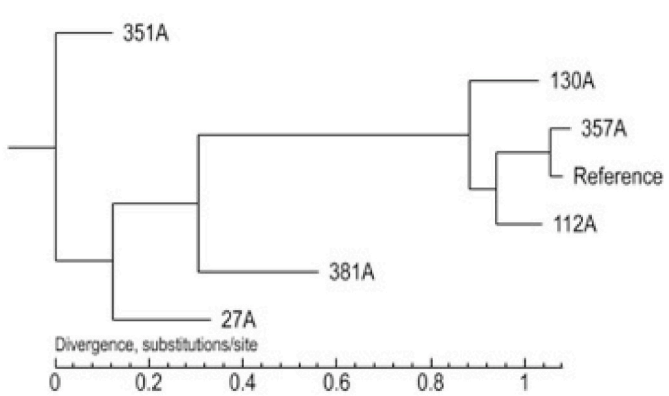


Fig. 4. System evolutionary tree of tested *E. coli* isolates. Reference: *E. coli* DH1, Sample: 112, 130, 357, 381, 351 and 27, based on SNPs analysis results.

Table 5The illumina sequencing data of MDR *E. coli* isolates.

Sample Name (#)	Total Reads (#)	Clean Data (Mb)	GC Content (%)	Coverage (%)	Total SNP	Insertion	Deletion	CDS InDel
27	4,386,314	570	50.42	87.84	48,023	127	115	42
112	4,386,332	571	50.45	92.55	14,269	51	49	27
130	4,432,952	574	50.74	91.11	19,485	57	68	24
351	4,339,668	571	50.41	89.01	48,030	139	126	45
357	4,433,010	572	50.42	95.37	3071	24	30	8
381	4,433,014	572	50.41	89	45,256	119	102	39

Table 6SNPs observed in MDR *E. coli* isolates with respect to the reference strain *E. coli* DH1.

Ref gene ID	Ref	<i>E. coli</i> 112	<i>E. coli</i> 130	<i>E. coli</i> 357	<i>E. coli</i> 381	<i>E. coli</i> 351	<i>E. coli</i> 27	Ref base <->Sample base	Ref codon<->Sample codon	Ref aa<->Sample aa	Mutate type	Ref gene product
EcDH1_1121	T	T	T	T	C	C	C	T<->C	TAT<->TAC	Y<->Y	syn	ABC transporter related protein
EcDH1_1231	T	T	T	T	C	C	C	T<->C	AAT<->AGT	N<->S	nonsyn	Beta-lactamase
EcDH1_1231	G	G	G	G	G	A	G	G<->A	AGC<->AGT	S<->S	syn	Beta-lactamase
EcDH1_0979	T	G	G	T	T	T	G	T<->G	GAA<->GAC	E<->D	nonsyn	Beta-lactamase domain protein
EcDH1_0979	T	C	C	T	T	C	T	T<->C	GTA<->GTG	V<->V	syn	Beta-lactamase domain protein
EcDH1_2586	A	A	A	A	C	C	C	A<->C	CAG<->CCG	Q<->P	nonsyn	Biofilm formation regulatory protein BssS
EcDH1_2806	G	G	G	G	A	A	G	G<->A	GCC<->GCT	A<->A	syn	Biofilm regulator BssR
EcDH1_2607	C	C	C	C	T	T	T	C<->T	GTC<->GTT	V<->V	syn	Curli assembly protein CsgF
EcDH1_2608	A	A	A	A	G	G	G	A<->G	AAA<->AAG	K<->K	syn	Curli production assembly/ transport component CsgG
EcDH1_1475	C	C	C	C	T	C	C	C<->T	CTC<->CTT	L<->L	syn	Drug resistance transporter Bcr/ CfIA subfamily
EcDH1_1582	T	T	G	T	G	G	G	T<->G	CTA<->CTC	L<->L	syn	Efflux transporter RND family
EcDH1_2144	A	A	T	A	A	A	A	A<->T	TAC<->TTC	Y<->F	nonsyn	MFP subunit FimH mannose- binding domain protein
EcDH1_3054	A	A	A	A	A	T	A	A<->T	GTG<->GAG	V<->E	nonsyn	Heavy metal efflux pump CzcA family
EcDH1_2117	T	C	T	T	C	T	T	T<->C	TTA<->CTA	L<->L	syn	Multiple antibiotic resistance (MarC)-related protein
EcDH1_2342	A	A	A	A	G	G	G	A<->G	ATT<->ATC	I<->I	syn	Phage shock protein A PspA
EcDH1_1400	G	G	A	G	G	G	G	G<->A	GCT<->GTT	A<->V	nonsyn	Small multidrug resistance protein

3.7. Identification of acquired resistance and virulence genes

It was found that there were total 7523 genes clusters were present in these isolates. Of them, 57 were involved in resistance against different antibiotics. Most of the resistant gene clusters have resistance genes for chloramphenicol, aminoglycoside, fluoroquinolone, beta-lactams, penicillin, cephalosporins, macrolides, tetracycline, resistance-nodulation-cell division (RND), multidrug resistance efflux pump (MDREP) and major facilitator superfamily transporter (MFST). Some important resistant gene clusters are given in Table 7.

Like resistance gene clusters, there were 601 virulence gene clusters.

Of them 552 were predicted from *E. coli* and other bacteria (of them 13 had unknown gene ID) and 49 were verified in bacteria including *E. coli*. Out of verified genes clusters 11 had their origin in *E. coli* while other 38 transferred from other bacteria. Some important gene clusters are given in Table 8. These genes were encoding lipopolysaccharide core biosynthesis protein, colonization factors, secretion systems, fimbrial chaperon and regulatory proteins, capsular polysaccharide synthesis enzyme, protein forming proteins and adhesin proteins which were directly involved in development of virulence.

The findings from ResFinder-4.1 database and VirulenceFinder-2.0 database, for the identification of antimicrobial resistance and

Table 7Important gene clusters having antibiotic resistance genes in MDR *E. coli* isolates.

Gene Cluster	<i>E. coli</i> 112	<i>E. coli</i> 130	<i>E. coli</i> 27	<i>E. coli</i> 351	<i>E. coli</i> 357	<i>E. coli</i> 381	ARDB: Identity	ARDB:E- value	Antibiotic resistance genes and drugs
[357] 357GL000608	0	0	0	0	100	0	100	2.00E-133	catA1-chloramphenicol
[112] 112GL000344	100	100	99.45	100	100	99.45	41.38	1.00E-11	catB- chloramphenicol
[27] 27GL001054	100	100	100	99.78	100	72.91	45.24	4.00E-99	oprM-aminoglycoside, tigecycline, fluoroquinolone, beta-lactam, tetracycline, RND and MDREP
[27] 27GL003390	0	0	100	100	0	0	45.8	3.00E-27	vatA,B-streptogramin, Virginiamycin, acetyltransferase.
[112] 112GL004503	100	99.74	99.49	0	0	0	47.59	3.00E-86	mexE,F-chloramphenicol, fluoroquinolone, RND, MDREP
[381] 381GL003916	99.53	99.41	99.64	99.65	99.53	100	53.44	0	pbp1A-penicillin
[112] 112GL004504	100	0	0	0	0	0	66.74	5.00E-157	ceoB- chloramphenicol, RND, MDREP
[112] 112GL002726	100	100	100	100	100	99.34	67.11	2.00E-61	tet3,4-tetracyclines
[112] 112GL002841	100	69.81	70.08	70.08	70.08	70.08	68	2.00E-147	acrA,B-aminoglycoside, glycylicycline, macrolide, beta-lactam, acriflavin, RND, MDREP
[27] 27GL004885	0	0	100	0	0	0	69.85	1.00E-55	fosA-fosfomycin
[112] 112GL000324	100	99.75	99.75	99.75	99.5	99.51	76.05	1.00E-169	rosA,B-fosmidomycin RosA: MFST, RosB: Potassium antiporter.
[27] 27GL004688	0	90.85	100	100	90.51	100	91.95	9.00E-164	tetC-tetracycline and tetracycline efflux pump.
[130] 130GL004349	0	100	99.5	99.5	0	99.5	100	0	cmIE3-chloramphenicol, MFST, chloramphenicol efflux pump.
[112] 112GL002575	100	100	98.67	98.45	97.08	100	100	0	bla1ec-cephalosporin, Class C beta-lactamase
[112] 112GL000341	70.97	100	100	99.9	70.97	99.9	100	0	acrA,B-aminoglycoside, glycylicycline, macrolide, beta-lactam, acriflavin, RND, MDREP
[112] 112GL001197	100	100	100	100	100	100	100	0	tolc, macA,B, mdtE,F, acrA,B- aminoglycoside, glycylicycline, macrolide, beta-lactam, acriflavin, RND, MDREP
[112] 112GL004511	100	100	94.16	100	100	80.07	100	2.00E-171	bl2be, ctxm-monobactam, penicillin, cephalosporins (all generations) ceftazidime, bla-a Class A beta-lactamase.
[112] 112GL003826	100	100	98.16	98.16	100	97.95	100	0	MdtP,N,O-acriflavine, puromycin, RND, MDREP
[112] 112GL003110	100	99.76	99.76	100	99.27	99.76	100	0	mdtM-chloramphenicol, acriflavine, norfloxacin, MFST, MDREP
[112] 112GL003190	100	100	99.49	99.74	100	100	100	0	mdtL-chloramphenicol MFST, MDREP
[112] 112GL003981	100	100	99.56	99.78	100	99.78	100	0	mdtK-enoxacin, norfloxacin, MFST
[112] 112GL001973	100	100	100	100	100	100	100	0	mdtG-deoxycholate, fosfomycin, MFST, MDREP

ARDB = antibiotic resistance genes database, Resistance-nodulation-cell division = RND, Multidrug resistance efflux pump = MDREP, Major facilitator superfamily transporter = MFST.

virulence genes are given in [Supplementary Tables S2 and S3](#), respectively.

4. Discussion

Our results about MDR isolates were supported by other researchers [66–68]. Owing to the importance of pathogenic colibacillosis in the poultry industry worldwide, antibiotics like fluoroquinolones, tetracyclines and cepheids were used to target *E. coli*. It clearly indicated an emergence of resistance of avian *E. coli* to these antibiotics [66]. Previously, Yang et al. demonstrated that more than 80% of *E. coli* collected from 10 poultry farms of China were resistant to at least eight antibiotics, including clavulanic acid, tetracycline and fluoroquinolones [67]. In our study, we found that a large numbers of *E. coli* isolates were resistant to these antibiotics. It was found from a study that isolates collected from poultry feces had a higher level of antimicrobial resistance than those isolates collected from poultry meat, beef and pork, indicating excessive use of antibiotics in poultry farming [69]. An epidemiological study from China was more consistent with our findings, which illustrated the data of two decades from 1993 to 2013, where 87.2% of poultry *E. coli* were MDR, mostly against five to six classes of antibiotics. Significant increase in

resistance was observed against ciprofloxacin, norfloxacin, levofloxacin, nalidixic acid, ampicillin, piperacillin, amikacin, chloramphenicol, trimethoprim-sulfamethoxazole, cefotaxime, ceftazidime and aztreonam in this study [70]. The results of Zhang et al. suggested that massive number of *E. coli* from septicemic broilers were MDR. The resistance of MDR was as; 97.3, 97.0, 95.5, 50.5 and 46.0% against streptomycin, kanamycin, gentamycin, neomycin and amikacin, respectively [71]. The results of recent study from China was in line with our findings, as they found that most of the poultry *E. coli* isolates were MDR (>92%) and high resistance was observed against aztreonam, gentamycin, penicillin, kanamycin, rifampicin and clindamycin [72]. In contrast to previous studies [70,71], pathogenic *E. coli* of poultry in the present study were mostly sensitive to tested antibiotics, from classes of penicillin, aminoglycosides, phenicol and phosphonic acid derivative. In Brazil, a high frequency of resistance against ampicillin, tetracycline, ceftazolin and nalidixic acid have found in commercially raised chicken, that is significantly higher than conventionally raised chicken, indicating the overuse of antibiotics on commercial farms [73]. Similarly, a high level of resistance was found against ciprofloxacin in poultry *E. coli* and isolates with >32 µg/mL MICs were found MDR and ESBLs positive [74].

Previously, it has also been noticed that patients diagnosed with

Table 8
Important virulence gene clusters of *E. coli* isolates.

Gene cluster	<i>E. coli</i> 112	<i>E. coli</i> 130	<i>E. coli</i> 27	<i>E. coli</i> 351	<i>E. coli</i> 357	<i>E. coli</i> 381	VFDB: Identity	VFDB:E- value	Virulence gene information
[112] _112GL003269	100	99.37	99.37	99.37	99.37	99.37	50.33	5E-42	{gb NP_208266 Verified (kdtB) lipopolysaccharide core biosynthesis protein LPS (VF0056)} [<i>Helicobacter pylori</i> 266951]
[112] _112GL001882	100	0	92.37	0	0	0	49.06	0	{gb NP_230493 Verified (actD) accessory colonization factor ActD [ACF (VF0127)] [<i>Vibrio cholerae</i> O1 biovar El Tor str. N16961]}
[130] _130GL002494	0	100	99.49	100	100	99.49	100	5E-107	{gb NP_286010 Verified (yagZ/ecpA) <i>E. coli</i> common pilus structural subunit EcpA [ECP (VF0404)] [<i>E. coli</i> O157:H7 str. EDL933]}
[351] _351GL002209	0	0	0	100	0	0	57.32	2E-48	{gb NP_462538 Verified (lpfD) long polar fimbrial protein LpfD [Lpf (VF0105)] [<i>Salmonella enterica</i> subsp. enterica serovar Typhimurium str. LT2]}
[351] _351GL003036	0	0	0	100	0	0	40.3	4E-09	{gb NP_541003 Verified (virB2) type IV secretion system protein VirB2 [VirB type IV secretion system (VF0365)] [<i>Brucella melitensis</i> 1 str. 16 M]}
[112] _112GL003137	100	100	100	100	100	100	100	9E-115	{gb NP_757240 Verified (fimE) Type 1 fimbriae Regulatory protein fimE [Type 1 fimbriae (VF0221)] [<i>E. coli</i> CFT073]}
[27] _27GL003421	99.33	99.67	100	100	99	99.33	99.34	2E-172	{gi 110644762 Predicted (fimH) FimH protein precursor [Type I fimbriae (CVF426)] [<i>E. coli</i> 536]}
[27] _27GL000440	0	83.12	100	100	84.62	100	100	1E-100	{unknown Predicted (ycbQ) fimbrial-like adhesin protein [<i>E. coli</i> laminin-binding fimbriae (ELF) (CVF824)] [<i>E. coli</i> O103:H2 str. 12009]}

MDR *E. coli* infection had a history of high consumption of poultry products [75]. Additionally, it has studied that there were genetic similarities between human and avian pathogenic *E. coli*, supporting the hypothesis that poultry and poultry products are notorious reservoir for MDR and pathogenic *E. coli* [12,76]. To conform the zoonotic threat of poultry *E. coli*, researchers found the co-presence of multi-locus sequence-type clones related to MDR in both poultry and human *E. coli* isolates [12,77], while the other researchers did not find such association [74].

ESBLs producing poultry *E. coli* in Henan province were 60.8% (31/51) [78]. which are comparable to the findings of our study. It was interesting to note that most of the checked β -lactamase genes were present in our selected pathogenic MDR *E. coli* isolates. a previous study found that fecal *E. coli* of healthy broilers also harbor β -lactamase genes [79]. To control the omphalitis in newly born chicken, it was practice to inject third generation cephalosporin, especially ceftiofur, in hatchable eggs. previously, it was assume that the cephalosporins might be the main contributor of an upsurge of β -lactamase genes in chickens [80]. However, broilers that had no history of cephalosporins treatment can also harbor β -lactamase genes [79]. Therefore, the assumption that β -lactamase genes are only because of ceftiofur treatment is not fully acceptable [79]. *E. coli* with β -lactamase genes were observed in wild animals, wild birds and in the environment [81].

ESBLs gene TEM was also found in selected isolates along with CTX-M genes, similar to results of previous study in chickens [82]. The CTX-M was the most common group of β -lactamase genes found from *E. coli* isolates of poultry [73,78], humans [83,84], retail food and river water [85]. CTX-M is derived from the favored activity for cefotaxime over ceftazidime [86]; therefore, higher resistance against cefotaxime was observed than ceftazidime in our study, might be owing to more prevalence of isolates harboring CTX-M genes [85]. The other most prevalent ESBL gene in our finding was CTX-M-1, which was found in other studies [12,13,87]. We also found TEM-1 in some isolates which already have been detected from poultry *E. coli* [85]. Our findings were in line with previous studies, as all selected MDR *E. coli* isolates were found with *pap-C*, *fim-C* and *fim-H* virulence genes. The pathogenic and virulence-associated gene *fim-C* was the most abundant gene isolated from the MDR *E. coli* of poultry in China [88]. These virulence-associated genes control a wide range of functions in adhesion, invasion, toxicity, immune suppression, iron acquisition and bacterial feasibility [24]. A case report showed that, most of detected virulent factors in our data were found in *E. coli* causing fatal necrotization in chicken [24]. Previously, *fim-C*, *iuc-D* and *iss* were the most prevalent virulence genes found in chickens and turkeys suffering from colibacillosis [89–91]. Presence of *iro-N* and *irp-2* in isolates supported the concept that iron acquisition systems play an important role in virulence [90].

Previously it has been studied that partially purified protein from cells expressing the mutant genes encoding these enzymes showed reaction rates higher than wild-type [92]. Similarly, salt tolerating mechanisms were also mutated in these isolates. As potassium efflux system protein and potassium antiporter were mutated. These mutated isolates can bear high osmotic pressure and high salt concentration, e.g. by accumulating high potassium from extracellular medium, *E. coli* can survive for long time [92]. Our RNA-Seq data also supports this statement, where salt controlling genes were up-regulated in *E. coli* isolate 381 [93]. The resequencing data showed that there were 163 gene loci encoding mutated numerous ATP-binding proteins, which have their targets for quinolones, chambered protease ClpP, acyl-co-A, ATP-synthase, carboxylase and aminoacyl-tRNA synthetase [94,95], ultimately leading to increase in resistance and virulence because ATP-binding protein targeted antibiotics will not work. ResFinder and VirFinder effectively-recognized every one of the genes from which the database was manufactured, and accurately distinguished all genes introduce in six *E. coli* isolates of entire genome information gathered from the NCBI genomes database (<http://www.ncbi.nlm.nih.gov/genome>). The

pan-genome alludes to the distinguishing proof of individual or populace particular successions that may contain imperative data relevant to the individual's uniqueness [94]. Besides, phenotypic antimicrobial resistance trial of six *E. coli* isolates were compared with the genome of reference strain in ResFinder. There were number of phenotypical and genotypical similarities found. Isolate 357 contain *catA*, which has 100% ARDB identity and *catB* gene was identified in all isolates, however, in phenotypical identification *catA* was identified in isolates 112 and 351. Our results were supported by a previous study where *catB9* gene was phenotypically silent in the normal state of bacteria [96]. Mutation in transcriptional regulator *MarR*, was present in all tested isolates, which cause resistance to quinolones, already studied in *E. coli* [97].

5. Conclusion

Antibiotics use in the poultry business and increase in poultry items utilization could have brought about the rise, spread, and ingenuity of antimicrobial resistance which is a genuine wellbeing sympathy towards both avian and human populations. Dispersal of MDR and pathogenic microbes worldwide is persistently diminishing the viability of accessible and commonly used antibiotics. Resequencing of MDR *E. coli* isolates showed dozens of resistance determinants in the genome. It may be responsible for the multidrug resistance of the isolates, although further confirmation studies are warranted. Numerous virulence factors identified in these isolates, suggesting that these isolates have pathogenic potential. The availability of the genome sequence and comparative analysis with *E. coli* DH1 strain provided the basis to further address the evolution of drug resistance mechanisms and pathogenicity in poultry *E. coli* isolates. Further investigations are required to find the correlation between virulence and resistance and the molecular mechanism of virulence and resistance, to develop an effective treatment against MDR poultry *E. coli*.

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CRedit authorship contribution statement

Hafiz Iftikhar Hussain: Writing – original draft, Methodology, Investigation. **Zahid Iqbal:** Conceptualization, Validation. **Mujahid Iqbal:** Validation, Formal analysis. **Xiuhua Kuang:** Methodology, Resources. **Yulian Wang:** Visualization, Resources. **Lingquan Yang:** Resources, Visualization. **Awais Ihsan:** Writing – review & editing, Validation. **Amjad Islam Aqib:** Formal analysis, Validation, Writing – review & editing. **Qari Muhammad Kaleem:** Validation, Software. **Yufeng Gu:** Data curation. **Haihong Hao:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.micpath.2022.105389>.

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